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PUBLICATION

00-README

Maven Smart System (MSS) — Training Curriculum

USAREUR-AF Operational Data Team

HEADQUARTERS
UNITED STATES ARMY EUROPE AND AFRICA
(USAREUR-AF)
Wiesbaden, Germany

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MAVEN SMART SYSTEM (MSS) — TRAINING CURRICULUM

USAREUR-AF OPERATIONAL DATA TEAM

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Distribution: Approved for public release; distribution is unlimited.

PURPOSE

This curriculum provides progressive, doctrine-aligned training for all USAREUR-AF personnel who access, build on, or lead with the Maven Smart System (MSS). It is organized into foundational doctrine publications and platform-specific technical manuals following Army publication conventions.

NEW HERE? START FAST

MSS Quick Start — Get operational in 30 minutes. Read this before anything else if you just got MSS access.

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HOW TO USE THIS CURRICULUM

Step 1 — Know Your Level

Level	You Are...	Start Here
All personnel	A Soldier, officer, or civilian who uses MSS to access operational data	TM-10
All staff, light builders	Building dashboards, forms, or basic pipelines on MSS	TM-20 (after TM-10)
Data-adjacent specialists	Assigned to a data/analytical role (17/25-series, S6/G6, G2/G9)	TM-30 (after TM-20)
Warfighting function staff	Assigned to INT, FIRES, M2, SUST, PROTECTION, or MC functions	TM-40A through TM-40F (WFF track)
Developers / technical specialists	Writing code, building external apps, ML models	TM-40G through TM-40L (technical track) after TM-30
Senior leaders (O-5+, CSM+)	Commanding or directing a data-capable formation	Data Literacy for Senior Leaders
All — background reading	Anyone who wants to understand data concepts before touching MSS	Data Literacy Technical Reference

TM-40 Series Disambiguation: The TM-40 series has two distinct sub-series: - **TM-40A through TM-40F** — Warfighting Function (WFF) tracks. Audience: functional staff in the six WFFs. Prerequisite: TM-20. These address how each WFF integrates MSS into its operational domain. - **TM-40G through TM-40L** — Technical Specialist tracks. Audience: engineers, analysts, and data professionals writing code or building models. Prerequisite: TM-30. These address platform development and advanced technical capabilities.

Do not assume a track based on letter alone — confirm against your role and the tables below.

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Step 2 — Read the Right Publications

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Senior Leaders ──────────────────────────► Data Literacy for Senior Leaders
                                                  │
                                                  ▼
All Personnel ───────────────────────────► Data Literacy Technical Reference
[recommended]
                                                  │
                                                  ▼
All Personnel ───────────────────────────► TM-10 (Maven User)
                                                  │
                                                  ▼
All Staff ───────────────────────────────► TM-20 (Builder)
                                            │            │
                                   ┌────────┘            └────────┐
                                   ▼                               ▼
                     WFF Staff (TM-40A–F)              Data-Adjacent Specialists
                     ────────────────────              ─────────────────────────
                     TM-40A (Intelligence)             TM-30 (Advanced Builder)
                     TM-40B (Fires)                            │
                     TM-40C (M2)                       ┌───────┴────────┐
                     TM-40D (Sustainment)              ▼                ▼
                     TM-40E (Protection)     TM-40G through TM-40L   TM-50 Series
                     TM-40F (Mission Cmd)    (Technical Specialists)  (Advanced)
```

Step 3 — Reference the Glossary

At any point, consult the Data & Foundry Glossary to translate between general data concepts and MSS/Foundry terminology.

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PUBLICATIONS INDEX

Doctrine Publications

Publication	File	Audience	Purpose
Data Literacy for Senior Leaders	doctrine/DATA_LITERACY_senior_leaders.md	O-5+, CSM+, Senior Civilians	Principles, command responsibilities, evaluating data products
Data Literacy Technical Reference	doctrine/DATA_LITERACY_technical_reference.md	All personnel	Comprehensive data literacy reference — platform-agnostic
Glossary — Data & Foundry Terms	doctrine/GLOSSARY_data_foundry.md	All personnel	Equates general data concepts to MSS/Foundry terminology

Technical Manuals (Platform-Specific)

Publication	File	Audience	Prerequisite
TM-10 — Maven User	tm/TM_10_maven_user/TM_10_MAVEN_USER.md	All staff	None
TM-20 — Builder	tm/TM_20_builder/TM_20_BUILDER.md	All staff	TM-10
TM-30 — Advanced Builder	tm/TM_30_advanced_builder/TM_30_ADVANCED_BUILDER.md	Data-adjacent specialists	TM-10, TM-20

TM-40A through TM-40F — Warfighting Function Tracks

These tracks address MSS integration within each warfighting function. Audience is functional staff assigned to that WFF. Prerequisite: TM-20.

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Publication	File	Concepts Guide	WFF	Prerequisite
TM-40A — Intelligence	tm/TM_40A_intelligence/TM_40A_INTELLIGENCE.md	CONCEPTS_GUIDE_TM40A	Intelligence (INT)	TM-20
TM-40B — Fires	tm/TM_40B_fires/TM_40B_FIRES.md	CONCEPTS_GUIDE_TM40B	Fires	TM-20
TM-40C — Movement & Maneuver	tm/TM_40C_movement_maneuver/TM_40C_MOVEMENT_MANEUVER.md	CONCEPTS_GUIDE_TM40C	Movement & Maneuver (M2)	TM-20
TM-40D — Sustainment	tm/TM_40D_sustainment/TM_40D_SUSTAINMENT.md	CONCEPTS_GUIDE_TM40D	Sustainment	TM-20
TM-40E — Protection	tm/TM_40E_protection/TM_40E_PROTECTION.md	CONCEPTS_GUIDE_TM40E	Protection	TM-20
TM-40F — Mission Command	tm/TM_40F_mission_command/TM_40F_MISSION_COMMAND.md	CONCEPTS_GUIDE_TM40F	Mission Command (MC)	TM-20

TM-40G through TM-40L — Technical Specialist Tracks

These tracks address advanced technical capabilities by specialty. Each includes a companion **Concepts Guide** covering background theory and doctrine context. Prerequisite: TM-30.

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Publication	File	Concepts Guide	Audience	Prerequisite
TM-40G — ORSA	tm/TM_40G_orsa/TM_40G_ORSA.md	CONCEPTS_GUIDE_TM40G	Operations Research analysts	TM-30
TM-40H — AI Engineer	tm/TM_40H_ai_engineer/TM_40H_AI_ENGINEER.md	CONCEPTS_GUIDE_TM40H	AI/ML specialists	TM-30
TM-40I — Machine Learning Engineer	tm/TM_40I_ml_engineer/TM_40I_ML_ENGINEER.md	CONCEPTS_GUIDE_TM40I	MLEs	TM-30
TM-40J — Program Manager	tm/TM_40J_program_manager/TM_40J_PROGRAM_MANAGER.md	CONCEPTS_GUIDE_TM40J	PMs, resource managers, G8/S8	TM-30
TM-40K — Knowledge Manager	tm/TM_40K_knowledge_manager/TM_40K_KNOWLEDGE_MANAGER.md	CONCEPTS_GUIDE_TM40K	KMOs, 37F, S2/S3/S6 KM roles	TM-30
TM-40L — Software Engineer	tm/TM_40L_software_engineer/TM_40L_SOFTWARE_ENGINEER.md	CONCEPTS_GUIDE_TM40L	SWEs	TM-30

TM-50 Series — Advanced Technical Specialist Tracks

Each TM-50 track builds directly on its TM-40G–L counterpart. Prerequisite is the corresponding TM-40G–L track.

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Publication	File	Concepts Guide	Audience	Prerequisite
TM-50G — Advanced ORSA	tm/TM_50G_orsa_advanced/TM_50G_ORSA_ADVANCED.md	CONCEPTS_GUIDE_TM50G	Senior ORSA practitioners	TM-40G
TM-50H — Advanced AI Engineer	tm/TM_50H_ai_engineer_advanced/TM_50H_AI_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50H	Senior AI engineers	TM-40H
TM-50I — Advanced MLE	tm/TM_50I_ml_engineer_advanced/TM_50I_ML_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50I	Senior ML engineers	TM-40I
TM-50J — Advanced Program Manager	tm/TM_50J_program_manager_advanced/TM_50J_PROGRAM_MANAGER_ADVANCED.md	CONCEPTS_GUIDE_TM50J	Senior tech PMs	TM-40J
TM-50K — Advanced Knowledge Manager	tm/TM_50K_knowledge_manager_advanced/TM_50K_KNOWLEDGE_MANAGER_ADVANCED.md	CONCEPTS_GUIDE_TM50K	Senior KMOs	TM-40K
TM-50L — Advanced Software Engineer	tm/TM_50L_software_engineer_advanced/TM_50L_SOFTWARE_ENGINEER_ADVANCED.md	CONCEPTS_GUIDE_TM50L	Senior SWEs	TM-40L

TRAINING LEVEL DESCRIPTIONS

TM-10 — Maven User (-10 Level)

Audience: All personnel **Competencies upon completion:** - Access MSS via CAC-based login - Navigate projects, datasets, and applications - Use Workshop applications to consume data and submit forms - Execute authorized Actions - Use Contour and Quiver for basic no-code analysis - Interact with AIP Logic and Agent interfaces - Understand classification markings and authorized export procedures - Troubleshoot common access and application issues

TM-20 — Builder (-20 Level)

Audience: All staff **Competencies upon completion:** - Create and organize Foundry projects via UI - Ingest data using Pipeline Builder (visual, no code) - Create basic Object Types and Link Types via Ontology Manager UI - Create simple Actions via Action Editor UI - Build and publish Workshop

applications with dashboards, forms, and filters - Manage project access and permissions via UI - Use Foundry branching to build and promote via UI - Follow USAREUR-AF naming conventions and builder standards

TM-30 — Advanced Builder (-30 Level)

Audience: Data-adjacent specialists (17/25-series, S6/G6, G2/G9, operational data analysts) **Competencies upon completion:** - Design and build complex multi-page Workshop applications with conditional logic and variable passing - Build advanced Pipeline Builder pipelines with multi-source joins and aggregations (visual) - Design Ontology structures (Object Types, Link Types, Actions) using the UI — architecture thinking, not coding - Conduct advanced analysis in Contour (complex aggregations, pivots, calculated columns, saved views) - Build advanced Quiver dashboards with multi-object analysis and linked views - Configure AIP Logic workflows (not author them) - Review and interpret data lineage graphs - Manage governance workflows with Data Stewards - Manage branching and production promotion via UI - Apply USAREUR-AF C2DAO governance standards and naming conventions

TM-40A through TM-40F — Warfighting Function Tracks

Each WFF track teaches how MSS capabilities are applied within that function's operational and analytical workflows. Prerequisite: TM-20 (not TM-30; these are functional user tracks, not coding tracks).

TM-40A — Intelligence: INT staff using MSS for collection management, targeting, and all-source analysis workflows.

TM-40B — Fires: Fires personnel using MSS for targeting data, effects assessment, and fire support coordination.

TM-40C — Movement & Maneuver: M2 staff using MSS for maneuver tracking, route analysis, and operational visualization.

TM-40D — Sustainment: G4/S4 and sustainment staff using MSS for logistics visibility, supply chain analytics, and readiness tracking.

TM-40E — Protection: Force protection and CBRN staff using MSS for threat tracking, vulnerability assessment, and protection COA support.

TM-40F — Mission Command: C2 and MC staff using MSS for operational dashboards, commander's critical information requirements (CCIR), and battle rhythm products.

TM-40G — ORSA (-40G Level)

Audience: Operations Research/Systems Analysis officers and NCOs, quantitative analysts **Competencies upon completion:** - Configure Code Workspaces (Python/R) within Foundry - Build statistical models: regression, classification, and validation for readiness/logistics applications - Conduct time series forecasting with ARIMA/SARIMA patterns - Run Monte Carlo simulation for COA comparison

and risk quantification - Apply linear programming for resource allocation and scheduling optimization - Design wargame/exercise data collection architecture and aggregation pipelines - Build analytical decision support products (Quiver/Contour) to commander standard - Communicate uncertainty: confidence intervals, sensitivity analysis, briefing standards

TM-40H — AI Engineer (-40H Level)

Audience: AI/ML specialists, data engineers assigned to AI pipeline development **Competencies upon completion:** - Author AIP Logic workflows: prompt engineering, chain design, output handling - Build and configure AIP Agent Studio agents with tools, memory, and orchestration - Implement LLM integration patterns: ontology data grounding, RAG, context construction - Apply AI safety requirements: human-in-the-loop gates, output validation, OPSEC for AI products - Write Python transforms that prepare data for AI consumption - Connect AIP Logic workflows to Object Types and Actions - Test and red-team AI outputs; evaluate quality against defined standards - Deploy and monitor AIP Logic workflows in production

TM-40I — Machine Learning Engineer (-40I Level)

Audience: ML engineers, data scientists building and deploying models on MSS **Competencies upon completion:** - Configure Code Workspaces for model development (GPU, packages, environment management) - Build and evaluate ML models within the Foundry environment - Manage model versioning, experiment tracking, and reproducibility - Deploy models to production and integrate with Ontology Objects and Actions - Implement MLOps patterns: monitoring, drift detection, retraining triggers - Apply responsible AI practices and model documentation standards for operational use

TM-40J — Program Manager (Technical) (-40J Level)

Audience: Technical project managers leading data, AI, and software capability builds on MSS

NOTE — Prerequisite Variance: The MTP lists TM-20 as the minimum prerequisite for TM-40J. The TM-40J syllabus lists TM-30. Commanders should route technical PMs who will oversee code-writing teams through TM-30 before TM-40J. PMs overseeing no-code or builder-level teams may proceed directly from TM-20.

Competencies upon completion: - Apply Agile project management (Scrum/Kanban) to data and AI projects - Write user stories and acceptance criteria for data products - Manage ML/AI project lifecycles from research through production - Translate between technical staff and operational users/commanders - Build sprint tracking and project status dashboards on MSS - Manage technical risk, dependencies, and delivery planning - Define production readiness and Definition of Done for data products - Lead change management and user adoption for new MSS capabilities

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TM-40K — Knowledge Manager (-40K Level)

Audience: KMOs, 37F, S2/S3/S6 knowledge management roles, unit S3/S6 knowledge officers

NOTE — Prerequisite Variance: The MTP lists TM-20 as the minimum prerequisite for TM-40K. The TM-40K syllabus lists TM-30. KMOs who will be building ingestion pipelines or AIP Logic workflows should complete TM-30 first. KMOs focused on taxonomy design and governance may proceed from TM-20.

Competencies upon completion: - Design knowledge architecture for AAR, lessons learned, doctrine, and SOP repositories on MSS - Build AAR capture systems using Workshop forms and Object Type pipelines - Design and operate lessons-learned ingestion and tagging pipelines - Use AIP Logic for knowledge summarization, search augmentation, and theme extraction - Build full-text and semantic search systems over knowledge repositories - Manage doctrine and SOP version control within Foundry - Build personnel expertise mapping (skills/experience registries) - Design knowledge transfer and unit continuity processes using MSS

TM-40L — Software Engineer (-40L Level)

Audience: Software engineers building external applications and platform integrations on MSS
Competencies upon completion: - Authenticate and query the Foundry Ontology via OSDK (TypeScript/Python) - Execute Actions, subscribe to Object changes, and handle pagination and filtering via OSDK - Use Foundry Platform SDK for dataset operations, file management, and branch management - Build TypeScript Functions on Objects (computed properties, bulk query patterns) - Write and test complex Action validators with TypeScript - Build Slate applications integrated with the Foundry API - Apply USAREUR-AF code review and deployment standards for MSS applications

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TRAINING MANAGEMENT

Administrative Publications

Resource	Location	Audience
Mission Training Plan (MTP)	training_management/MTP_MSS.md	Training managers, S3/G6, commanders
Program of Instruction (POI)	training_management/POI_MSS.md	Training managers, curriculum designers
Course Administrative Data (CAD)	training_management/CAD_MSS.md	Training administrators
Training and Evaluation Outline (TEO)	training_management/TEO_MSS.md	Instructors, evaluators
Annual Training Schedule	training_management/ANNUAL_TRAINING_SCHEDULE.md	Training managers, S3
Enrollment SOP	training_management/ENROLLMENT_SOP.md	Training coordinators
Completion Certificate Template	training_management/COMPLETION_CERTIFICATE.md	Training administrators
Faculty Development Plan	training_management/FACULTY_DEVELOPMENT_PLAN.md	Instructor cadre, S3
Training Policy Letter	training_management/POLICY_LETTER.md	Commanders, training managers
Curriculum Maintenance SOP	training_management/CURRICULUM_MAINTENANCE_SOP.md	Curriculum owner, SMEs, instructors
AAR / Feedback Template	training_management/AAR_TEMPLATE.md	Instructors, evaluators

The MTP provides: TLOs/ELOs per level, individual task lists, Go/No-Go criteria, hour-by-hour blocks of instruction for all levels, training schedule templates, resource requirements, instructor certification standards, practical exercise scenarios (all levels including TM-40), and sustainment requirements.

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Lesson Plans

Resource	Location	Notes
Lesson Plan Template	training_management/lesson_plans/LP_TEMPLATE.md	Blank template for new courses
TM-20 Lesson Plan Outlines	training_management/lesson_plans/TM20_LESSON_PLAN_OUTLINES.md	Builder track
TM-30 Lesson Plan Outlines	training_management/lesson_plans/TM30_LESSON_PLAN_OUTLINES.md	Advanced Builder track
TM-40 Specialist Lesson Plan Outlines	training_management/lesson_plans/TM40_SPECIALIST_LESSON_PLAN_OUTLINES.md	TM-40G through TM-40L technical tracks

Course Syllabi (Student/Instructor-Facing)

Hand the appropriate syllabus to each trainee at the start of their course.

Syllabus	File	Level
TM-10 — Maven User	syllabi/SYLLABUS_TM10.md	All personnel
TM-20 — Builder	syllabi/SYLLABUS_TM20.md	All staff
TM-30 — Advanced Builder	syllabi/SYLLABUS_TM30.md	Data-adjacent specialists
TM-40G — ORSA	syllabi/SYLLABUS_TM40G.md	ORSA analysts
TM-40H — AI Engineer	syllabi/SYLLABUS_TM40H.md	AI/ML specialists
TM-40I — ML Engineer	syllabi/SYLLABUS_TM40I.md	MLEs
TM-40J — Program Manager	syllabi/SYLLABUS_TM40J.md	PMs, G8/S8
TM-40K — Knowledge Manager	syllabi/SYLLABUS_TM40K.md	KMOs, 37F
TM-40L — Software Engineer	syllabi/SYLLABUS_TM40L.md	SWEs

NOTE

WFF track syllabi (TM-40A through TM-40F) and TM-50 series syllabi are not yet published. WFF track instruction is managed through the MTP and TEO. TM-50 advanced track instruction is managed through the MTP. Contact training management for scheduling.

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SUPPORTING REFERENCE MATERIAL

Resource	Location	Notes
MSS Quick Start	QUICK_START.md	30-minute onboarding for new MSS users
Naming & Governance Standards	standards/NAMING_AND_GOVERNANCE_STANDARDS.md	USAREUR-AF MSS naming conventions, governance checklists
MSS Training Hub (web)	mss_info_app/index.html	Standalone HTML training hub — open in browser
Practical Exercises	exercises/	Hands-on exercise packages for TM-10 through TM-40L
Foundry Python Patterns (runnable code)	data_skills/13_foundry_patterns/	Local shim for offline development and testing
Full data science reference library	data_skills/	15 modules: Python → ML → ETL → deployment
SITREP Tracker (applied project)	sitrep_tracker/	Click + Rich + SQLite — reference implementation

QUICK-REFERENCE TRAINING PATHS

For a New Soldier joining the data team:

1. Read Data Literacy for Senior Leaders if O-5+; skip if not
2. Skim Data Literacy Technical Reference Ch 1-3 and Ch 8
3. Complete TM-10 cover to cover
4. Request MSS access; practice navigation tasks
5. If building: complete TM-20
6. If WFF functional staff: proceed to the appropriate TM-40A–F WFF track after TM-20
7. If data specialist: complete TM-30 after TM-20
8. If technical specialist: proceed to the appropriate TM-40G–L track after TM-30

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For an S6 shop standing up MSS capability:

1. Commander reads Data Literacy for Senior Leaders
2. All staff complete TM-10
3. Designate builders: complete TM-20
4. Designate data leads: complete TM-30
5. Reference Glossary throughout
6. Technical specialists proceed to TM-40G–L tracks per role assignment
7. WFF functional staff proceed to TM-40A–F track per function

By MOS/Role — TM-40 Track Selection:

Role / MOS	Recommended TM-40 Track
All WFF staff	TM-40A through TM-40F per assigned warfighting function
17A/17C — Cyber officer/NCO	TM-40L (SWE) or TM-40H (AI Eng)
25D — IT specialist	TM-40L (SWE)
25U — Signal support	TM-40L (SWE) or TM-40J (PM)
FA49 — Operations Research	TM-40G (ORSA)
37F — Psychological Operations	TM-40K (KM)
G2/S2 analyst	TM-40G (ORSA) or TM-40K (KM)
G6/S6 data officer	TM-40L (SWE) or TM-40H (AI Eng)
G8/S8 resource manager	TM-40J (PM)
G9/Civil Affairs	TM-40J (PM) or TM-40K (KM)
Data scientist (GS/contractor)	TM-40G (ORSA) or TM-40I (MLE)
AI/ML engineer (GS/contractor)	TM-40H (AI Eng) or TM-40I (MLE)
KMO / Knowledge Officer	TM-40K (KM)
PM / Program Manager	TM-40J (PM)

USAREUR-AF Operational Data Team — UNCLASSIFIED Version 5.0 | March 2026 | TM-40 WFF and Technical Specialist tracks complete; TM-50 series complete | Supersedes v4.0

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